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标题	N - Mesyl-Enabled Cu2O Catalysis: Synthesis of (E) - 3-Alkylideneisoindolin-1-ones and 3,4-Unsubstituted Isoquinolones via Sequential Alkynylation/Annulation
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摘要（原文节选）

A highly regio- and stereoselective synthesis of (E)-3-alkylideneisoindolin-1-ones has been developed via a Cu2O-catalyzed ligand- and base-free Sonogashira alkynylation/5-exo-dig cyclization of o-iodo-N-mesylbenzamides with terminal alkynes. In this process, the N-mesyl group serves as a dual directing and activating group, ensuring the exclusive formation of the (E)-exocyclic double bond in good-to-excellent yields with broad functional group tolerance. The mesyl group can be subsequently removed with TBAF to aff ⋯

数据表 · 1（19 行，含数值行 19 行）

Screening of the Reaction Conditions for Sonogashira Alkynylation

entry	catalyst	solvent	time (h)	temperature (° C)	%conversion
1	Cu2O	DMF	8	90	14
2	Cu2O	DMF	12	90	43
3	Cu2O	DMF	24	90	61
4	Cu2O	DMF	8	105	70
5	Cu2O	DMF	12	105	80
6	CuCl	DMF	12	105	N.R
7	CuBr	DMF	12	105	N.R.
8	CuI	DMF	12	105	N.R.
9	Cu(OAc)2	DMF	12	105	N.R.
10	CuSO4	DMF	12	105	N.R.
11	Cu(OTf)2	DMF	12	105	N.R.
12	Cu	DMF	12	105	N.R.
13	Cu2O	MeOH	12	105	N.R.
14	Cu2O	t-BuOH	12	105	23
15	Cu2O	THF	12	105	17
16	Cu2O	dioxane	12	105	30
17	Cu2O	DME	12	105	29
18	Cu2O	MeCN	12	105	23
19	Cu2O	PhMe	12	105	11

脚注（逐字）：aReactions were performed on a 0.1 mmol scale with the indicated catalyst (30 mol %) and solvent (0.5 mL) in sealed pressure-relief borosilicate glass vials.bYields of conversion were based on HPLC.cNo reaction.

数据表 · 2（9 行，含数值行 9 行）

Coupling of Diverse o-Iodoaryl Carboxamides with Terminal Alkynes under Cu2O-Catalyzed Conditions

entry	subs./prod.	R1	R2	yield (%)
1	1a/2a	– H	– CO2Et	79
2	1b/2b	– Cl	77	
3	1c/2c	– Br	71	
4	1d/2d	– NO2	79	
5	1e/2e	– CH3	71	
6	1a/2f	– H	– Ph	75
7	1c/2g	– Br	65	
8	1a/2h	– H	– TIPS	74
9	1d/2i	– NO2	70	

脚注（逐字）：aReactions were carried out in pressure-relief borosilicate glass vials.bIsolated yields.

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